

BrickAtlas: From Valid Brick Structures to Constructible and Recoverable Processes

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Author and affiliation metadata intentionally omitted

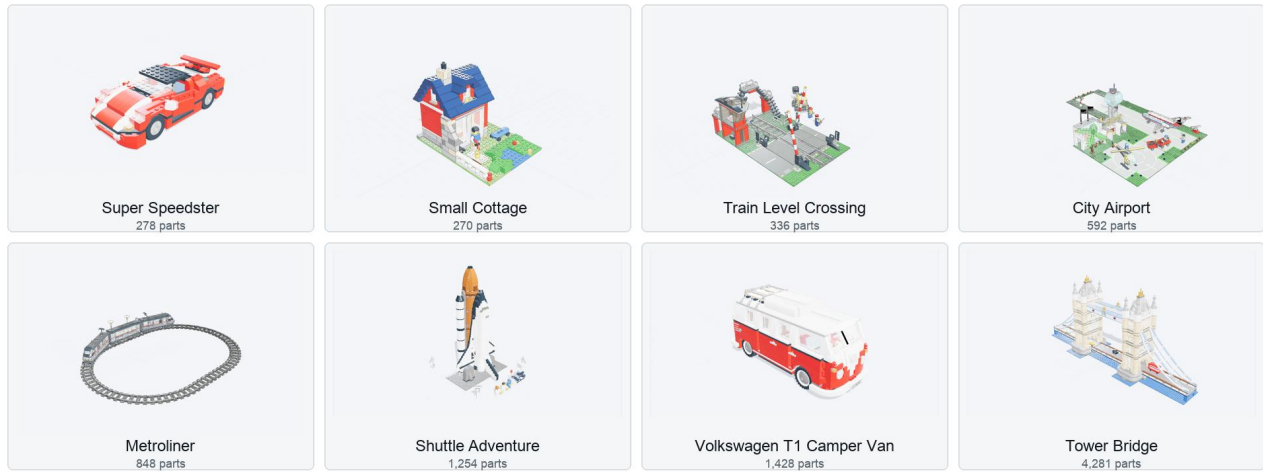


Figure 1. **Eight of 15 primary OMR/LDraw objects.** The licensed whole-model registry spans 36–4,281 placed instances, 624 part numbers and 42 colors. Thirty-nine source-named semantic subassemblies provide coherent components such as train cars, aircraft, bridge towers, decks, roofs and cables. Random-growth structures are retained only for historical mechanism regression.

Abstract

A valid final brick structure does not certify a constructible process: an intermediate state may be unsupported, a placement may leave no tool clearance, a local error may require rollback, or missing evidence may require clarification rather than a guess. BrickAtlas separates these failures with explicit information and action contracts. Its primary object registry contains 15 licensed OMR/LDraw whole models and 39 source-named semantic subassemblies, totaling 9,935 placements across 13 set families. Six original 158–408-part objects support controlled process protocols. The new Constructibility-1 release provides 78 sequence-audit, blocked-recovery, stockout-replanning, and calibrated-clarification conditions. A body-only insertion rule misses all six designed clearance dead ends. Public-information solvers pass all conditions, while named shortcuts pass only 6/18, 0/18, 0/6, and 5/36 conditions respectively. Diagnostic-2 adds 138 paired diagnosis–access–repair conditions, and a finite grammar supplies 600 ambiguity conditions with exhaustive answer sets. We retain raw outputs, source groups, costs, and replayable evaluation ev-

idence. Constructibility-1 uses nominal support and clearance proxies and has no new learned-model, human, physical, or independent-source results; those remain required before confirmatory claims.

1. Introduction

A system can recognize the bricks in an object without recovering their poses, match its exterior without identifying hidden connections, or return a nearly unchanged structure with high similarity after failing an edit. A correct final assembly also need not make a submitted construction order executable. Treating these outcomes as a single notion of “structural intelligence” obscures what a model accomplished.

BrickAtlas uses shared source assemblies to separate these endpoints. Figure 3 shows actual inputs and ground truth, while Fig. 5 distinguishes a valid sequence from a dead end. Following dataset/model reporting and holistic evaluation principles [5, 15, 22], we separate information, outputs, equivalences and efficiency.

This is not a claim to originate brick generation, re-

construction, or multi-model assembly evaluation. BrickGPT [25] and BrickNet [14] provide strong generative foundations; MEPNet [31], Neural Assembler [32], and TreeSBA [8] address visual assembly. BC-Bench [16] evaluates selection and pose; PhyBlock [18] covers diagnosis and planning robustness. Our narrower opportunity is *measurement validity across shared objects*: when does a plausible appearance, a high similarity score, or one valid reference justify a downstream structural conclusion?

Our central question is whether a system can produce a process whose prefixes remain geometrically legal, supported, accessible, and recoverable, while requesting information when evidence is insufficient. We represent this as the diagnostic vector $\mathcal{C} = (G, P, A, R, U)$ rather than a weighted score that lets static geometry compensate for failed recovery. Diagnostic-2 separately tests diagnosis–access–repair coherence. Historical studies examine global versus changed-region similarity, hidden-relation identifiability, and candidate-order sensitivity.

The release contributes (1) a licensed whole-model and semantic-subassembly registry instead of random-growth publication examples; (2) a process-aware constructibility contract joining prefix validity, tool clearance, minimum recovery, finite-stock replanning and calibrated inspection; and (3) source-grouped shortcut audits, paired controls and raw-output replay. We separate primary objects, controlled mechanism probes, historical diagnostics and unrun experiments throughout.

2. Related Work and Position

Generation and assembly. BrickGPT [25] generates stable and buildable objects from text. BrECS [1] studies budget-aware generation and completion with efficient constraint checking. BrickNet [14] uses human-designed LDraw objects and typed connectors; its 320,808 samples and 9,743 part variants far exceed our grid vocabulary. Nominal contact in BrickAtlas is not a substitute for their structural or physical scope. MEPNet [31] reads stepwise manuals; Neural Assembler [32] predicts components, topology, poses, and sequence from multiple views; TreeSBA [8] uses a tree representation and synthetic-to-real supervision. Break and Make [30] studies interactive inspection, disassembly, and rebuilding. Their input protocols differ from fixed image or symbolic submissions and require audited adapters for comparison.

The closest multimodal benchmark. BC-Bench/Brick-Composer [16] evaluates diverse-brick selection and pose estimation from manual context and current state, including feedback and training. LEGO-Puzzles [29] also includes multi-step plans; PhyBlock [18] studies progressive planning, diagnosis and counterfactuals. Co-builder [10] addresses state detection, and WorkBenchMark [19] includes

robot/VLA assembly. These are not empty research areas. Our bounded distinction is a shared-object conjunction of per-prefix support, explicit tool clearance, costed local recovery, stockout continuation and observation-consistent uncertainty. It is not a claim of first-ever assembly reasoning or robot execution.

Diagnostic visual reasoning. CLEVR [13] establishes controlled compositional reasoning evaluation; BLINK [4] tests perceptual skills, including multi-view and spatial reasoning. CSGNet [28] predicts executable shape programs. SpatialBabel [17] examines reconstruction across scene-code languages. Output-protocol effects are therefore not new in themselves. Answer consistency also has prior work [11, 12]; our distinction concerns executable maintenance constraints. Similarly, MCQ position bias is established [33]; our candidate experiment specifically compares actual pose changes against an identical-repeat baseline, without inferring internal model mechanisms from external API replies.

3. Benchmark Construction and Tasks

3.1. Geometry, Sources, and Splits

The primary source layer contains 15 complete OMR/LDraw files already used by the BrickAtlas application. They preserve source hierarchy, part identities, colors, arbitrary LDraw transforms, author, license, URL and source hash. Complete models contain 36–4,281 instances. We select 39 source-named semantic descendant closures and exclude random crops, step fragments, minifig-only cases, lamps and cones. All cases from a model retain one source group; alternate builds from the same set retain one set family.

Controlled process tasks use a separate nominal grid. There a part is (i, t, c, x, y, z, r) : instance ID, type, color, integer minimum corner and quarter-turn yaw. Geometry requires no overlap and at least one directly supporting stud for elevated parts. Force balance, clutch, arbitrary CAD contact and robotic grasping remain outside that contract.

The earlier 5,120 random-growth structures and their 117,910 derived conditions are excluded from current object scale and publication-facing galleries. They remain only to reproduce historical model runs, scorer regressions and shortcut censuses. More views, tasks and subassemblies never create additional independent objects.

Curated and challenge layers. The twelve 8–28-part curated objects and six 49–63-part Challenge-2 objects remain calibration layers. Frontier-1 (Fig. 1) adds six 158–408-part objects, 1,509 placed parts, 14–24-stud depth, and dependency chains up to 19. Its separate 512-part, 96-cell envelope does not change historical limits. Eight tasks per ob-

Work	Supplied information	Main output / setting	BrickAtlas distinction
BrickGPT; BrECS	Text or target constraints	Generation / completion	Multi-task measurement, not new generation
BrickNet	LDraw parts and typed connectors	Graph-backed build programs	Smaller domain; explicit answer equivalence
MEPNet; TreeSBA	Manuals or multi-view images	Poses / assembly actions	Shared task labels and changed-region metrics
Break and Make	Interactive observations	Disassembly and rebuilding	Fixed-input and tool tracks kept separate
BC-Bench	Manual, current state, candidate bricks	Selection and single-step pose	Hidden-layout ambiguity and typed multi-task success
SpatialBabel	Images and scene-code languages	Primitive scenes / spatial QA	Nominal contacts, edits, and original-order execution

Table 1. Closest-work positioning by input and output protocol. This is not a capability checklist or an empirical comparison on matched data. Specialized-method adaptation remains unrun.

Layer	Groups	Records
OMR whole/semantic	15	54
Frontier process	6	264
Finite ambiguity	9	600
Procedural v2 (retired)	5,120	117,910

Table 2. Dataset layers are not pooled. OMR has 13 set families; Frontier’s 264 records combine 48 Frontier-1, 138 Diagnostic-2 and 78 Constructibility-1 conditions on the same six sources.

ject cover minimum-action service, access certificates, parallel scheduling, minimum support intervention, inventory replacement, hidden seam sets, contingent inspection and mixed-fault repair. An audit found identical answers across all six legacy ambiguity and inspection cases. Diagnostic-2 replaces these templates and adds 72 paired maintenance conditions, 36 geometry-observation conditions, 18 deadline schedules and 12 mixed-pose conditions. Full references are withheld from the layer-image track. The supplement defines all new contracts. Constructibility-1 adds 78 conditions over the same six public sources: 18 immutable-sequence audits, 18 blocked recoveries, six in-process stock-outs, and 36 ambiguous/resolved clarification decisions. These are derived development conditions, not independent objects or model results.

3.2. Task and Information Contract

Table 3 defines the task families. **Recognition** predicts a known catalog type, color, and stud count from isolated views. **Relations** uses a symbolic assembly and two IDs to ask direct contact, vertical ordering, mating-stud count, and shortest undirected contact-graph distance; a non-contact pair can still have a graph path. **Reconstruction** predicts a full structure from reference information and BOM. **Generation** accepts any legal structure satisfying exact extents,

zero anchoring, a piece budget, color count, connected component count, and minimum maximum contact degree. It is constrained layout generation, not semantic text-to-shape evaluation.

Completion removes the last $\lfloor n/4 \rfloor$ pieces of a valid insertion sequence and requires preserving the remaining prefix. It does not cover arbitrary holes. **Editing** rotates the whole structure, replaces all occurrences of one color, or removes one topmost piece. Rotation maps positive X toward positive Z , keeps Y , and reanchors minimum X, Z . **Repair** has no-fault, color-fault, and one-stud-shift cases on exposed topmost pieces. It requires corrected geometry and faulty current IDs, with an empty fault list for a normal input. **Planning** supplies the target with IDs in lexical order and requests any executable permutation for vertical assembly/disassembly.

Each source yields 13 variants and 23 information-conditioned cases. Reconstruction, completion, and all three repairs have three reference conditions. With 25 catalog types in six colors, the total is $5,120 \times 23 + 150 = 117,910$. **Ordinary RGB** provides isometric/top/front/side assembled views and BOM; interiors remain hidden. **Layer disclosure** provides an assembled view and top images of each bottom- Y layer with other layers removed. **Symbolic reference** supplies complete geometry. The latter two add information and require full type/color/pose recovery. Other families use their declared symbolic or constraint inputs. Fault labels, changed IDs, and private render recipes are excluded from ordinary prompts.

4. Ground Truth and Measurement

4.1. Visual Compatibility Versus Exact Structure

Ground-truth records bind input hashes, source groups, target/source geometry, protected and changed IDs, contacts, nominal stud mates, executable witnesses, and metric appli-

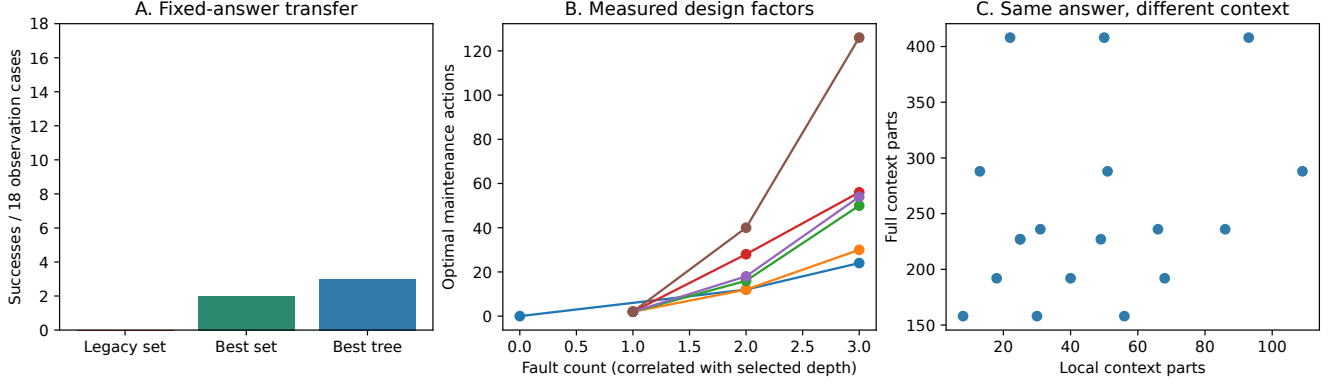


Figure 2. Diagnostic-2 design audits, not learned-model performance. Fixed-answer transfer uses shared candidate IDs; the best released set and tree transfer to 2/18 and 3/18 cases. Matched contexts have identical answers despite different total sizes. Fault count and selected depth remain correlated; these are not fully orthogonal difficulty strata.

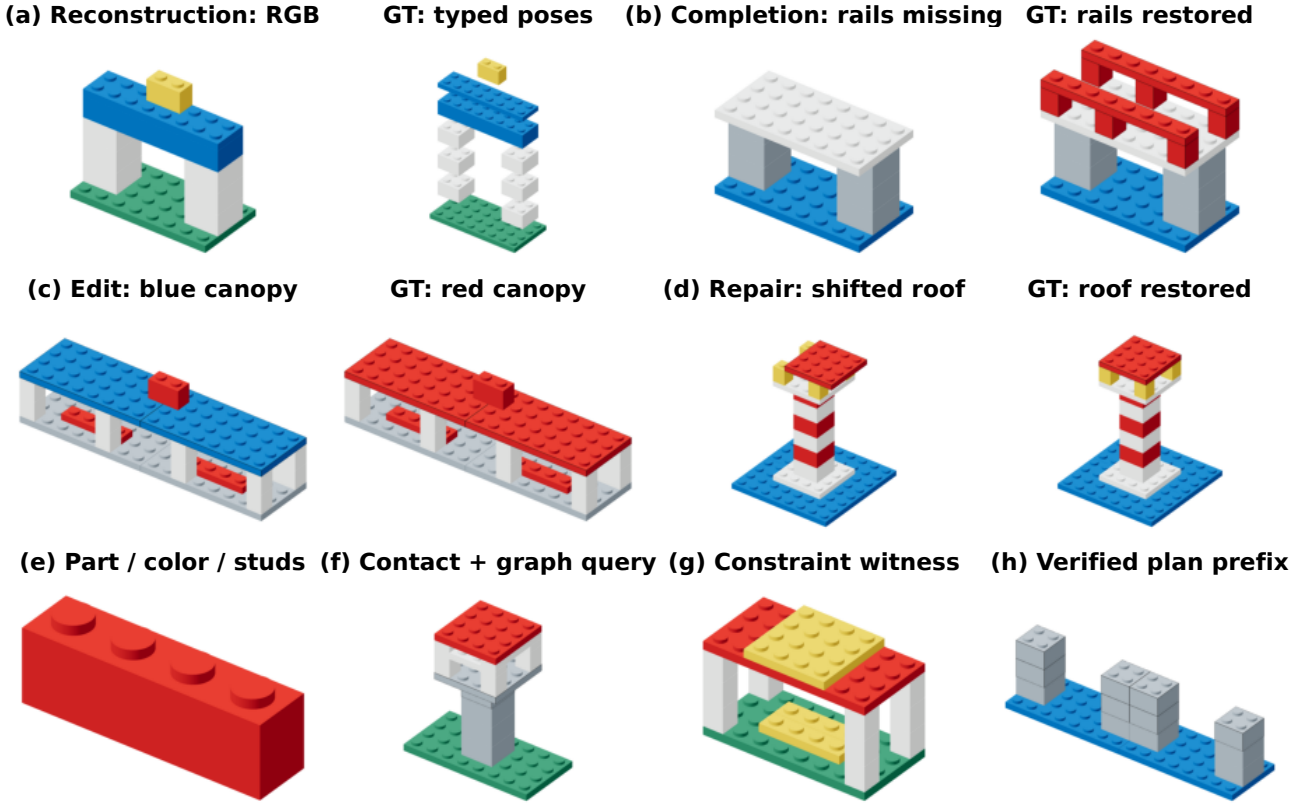


Figure 3. **Task-specific curated input and ground truth.** Reconstruction maps gate views to typed poses; bridge completion restores rails; station editing changes only the canopy color; lighthouse repair restores the shifted roof. Bottom panels illustrate part recognition, a typed relation query, one valid generation witness, and a verified planning prefix. Exploded offsets are display-only, not valid poses or ordinary model observations.

cability. Multiset matching ignores arbitrary output IDs and symmetric per-part yaw, but does not silently align global translation or rotation. For m matched parts in target T and

answer A ,

$$P = m/|A|, \quad R = m/|T|, \quad F_1 = 2m/(|A| + |T|). \quad (1)$$

Family (variants)	Public input	Output	Success contract	Cases
Part recognition	2 isolated RGB views	Type, color, stud count	All three fields correct	150
Relations (2)	Symbolic structure + two IDs	Connection, above, studs, distance	All four fields correct	10,240
Reconstruction	Reference + BOM	Full structure	Condition-specific match + legality	15,360
Generation	Six public constraints	Any satisfying structure	All constraints + legality	5,120
Completion	Current prefix + reference + BOM	Full structure	Match + preserved prefix	15,360
Editing (3)	Current structure + operation	Full edited structure	Exact edit + protected preservation	15,360
Planning (2)	Target structure + direction	Permutation of all IDs	Complete executable order	10,240
Repair (3)	Current + reference + BOM	Structure + faulty current IDs	Match + preservation + localization	46,080

Table 3. Eight task families. “Reference” denotes the declared RGB, layer, or symbolic condition. Symbolic tasks are not visual scores.

BOM F1 removes pose; occupancy IoU removes part boundaries; edge F1 requires matched endpoints and mating-stud counts.

For ordinary reconstruction, let S^+ be colored first-hit cells along positive X, Y, Z . We accept

$$V(A, T) = \mathbf{1}\{\text{valid}(A)\} \mathbf{1}\{\text{BOM}(A) = \text{BOM}(T)\} \cdot \mathbf{1}\{S^+(A) = S^+(T)\}. \quad (2)$$

This proxy allows alternative hidden arrangements; it is not arbitrary camera visibility or pixel similarity. Full-reference correctness is reported separately. Completion subtracts preserved source matches before computing *addition* F1. Edits/repairs require protected piece preservation; repairs additionally require faulty-ID localization. False alarms are defined only for no-fault cases.

RQ1: no-operation shortcuts. Copying a prefix with k missing parts from an n -part target gives

$$F_1^{\text{copy}} = \frac{2(n-k)}{2n-k}, \quad F_{1,\text{added}}^{\text{copy}} = 0. \quad (3)$$

Copying before a one-piece removal similarly yields $2(n-1)/(2n-1)$, approaching one without performing the edit. A full 35,840-case no-operation census confirms this: 87.1% completion and 97.4% removal F1, both with zero task success. Positive repair copies reach 94.9% F1 without correct localization. Normal no-fault repair correctly succeeds unchanged. All seven endpoints contain one case per source; per-case results and complete global/local metrics remain available.

4.2. An Ambiguity-Aware Extension

Matching appearance need not identify hidden relations. Instead of forcing one reference answer, we make the admissible set explicit. The new, separately versioned track uses

Condition	Groups	Cases	Yes	No	Undet.
Exterior	9	300	0	210	90
Support disclosed	9	300	36	264	0

Table 4. New finite-grammar task counts, separate from v2. Nine parameterized families are not nine independent semantic sources.

sealed shells with a $2 \times W$ internal layer, $W \in \{2, 4, 6\}$, tiled by identical 1-by-2 bricks. Three stack heights give nine parameterized families. The tilings are exhaustive within this finite grammar: their counts are 2, 5, and 13, independently checked by the recurrence $c_W = c_{W-1} + c_{W-2}$, $c_0 = c_1 = 1$. There are 60 valid layout states across the nine families.

For each pair of blue upper bricks we ask whether they share a directly supporting red brick. Given the observation O and public grammar $\mathcal{G}$, let $\mathcal{A}(O, \mathcal{G})$ be all admissible layouts. With Boolean query q ,

$$y(O) = \begin{cases} \text{yes}, & \forall A \in \mathcal{A}, q(A) = 1, \\ \text{no}, & \forall A \in \mathcal{A}, q(A) = 0, \\ \text{undetermined}, & \text{otherwise.} \end{cases} \quad (4)$$

Exterior and support-disclosed conditions create 600 paired cases (Table 4). Ninety exterior questions are genuinely ambiguous within the declared grammar; disclosure resolves them. Known geometry and the grammar are supplied symbolically, so this extension tests observation-consistent structural inference, not pure visual perception. Public records omit labels and hidden supports; an enumeration solver consumes only these public inputs and solves all 600 cases.

All layouts agree in BOM, colored occupancy, and declared surfaces and pass two geometry implementations. A separate high-resolution renderer checks all 60 layouts in

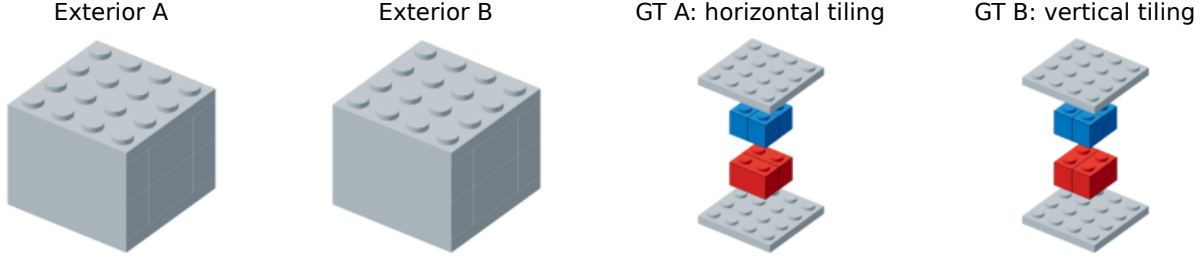


Figure 4. **Same exterior, different ground truth.** The two sealed structures have identical rendered exterior pixels under the tested cameras. Exploding their layers reveals horizontal versus vertical red internal bricks; side walls are omitted in the disclosed views. The outlined adjacent blue pair shares one red supporter in A but not B. The exterior-only answer is therefore *undetermined*; disclosing the support layer makes the answer definite. Exploded views are evaluator illustrations.

four views: 240 rendered frames match their within-family reference pixel-for-pixel. This result is renderer/camera-specific, not human calibration. Unweighted always-no accuracy is 70% exterior and 88% disclosed; class-balanced accuracy is 50% in both. We therefore report balanced success, false certainty on ambiguous cases, and unnecessary abstention on determinate cases. Complete baseline and planned VLM tables are in the supplement.

4.3. Execution and Evaluation Integrity

Execution stops at the first unknown, duplicate, unsupported, colliding, or blocked action. We record legal-prefix fraction, coverage, duplicate status, and the zero-based failure step. An incomplete legal prefix fails success but has no illegal step. Height sorting solves all 10,240 plans in this vertical domain: supporters are lower and already present; an earlier part overlapping the footprint must end below the new part, or the final geometry would collide. Reversing the sequence gives disassembly. This establishes a constraint-following ceiling, not difficult search.

Frozen selection manifests reject unknown/duplicate predictions; missing/malformed submissions remain in applicable denominators. Null means inapplicable, not a removed failure. A versioned strict decoder closes legacy primitive-type coercion without altering recorded scores. All 117,910 retired-regression oracle answers pass; independent metrics and mutation checks are detailed in the supplement. Replay labels model, harness, environment, and evaluator operations separately. Visual unfolding of an unordered structure is never called a model plan.

5. Models, Baselines, and Experiments

5.1. Historical Eight-Model Diagnostic

For reproducibility only, we retain the same 39 task cases from three retired procedural objects (12/20/29 parts) with eight models [2, 7, 21, 23, 27]. Four models retain their original verified responses; four new models receive iden-

tical prompts, image bytes, and the 2,200-token JSON output cap. These scores are not results on the primary OMR registry. Temperature is zero, model order rotates by case, automatic retries and provider fallback are disabled, and provider, tokens, truncation, latency, and settled OpenRouter charges [24] are retained. Gemini reasoning is disabled; other models have no requested reasoning budget. Cross-round time/provider differences remain potential confounds.

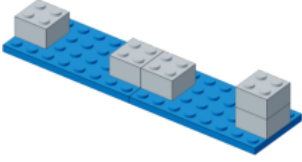
The supplement separates all 13 variants and same-source controls: copying, train-only retrieval, symbolic rules and height sorting. Unsupported pairs are not failures. Table 6 separates format, local changes and exact success; scores do not identify internal perceptual or reasoning mechanisms. In this shared sample, 3 of 13 variants have zero exact success across all eight models and 0 are saturated. Reconstruction successes range from 0/3 to 0/3; mean added-part F1 ranges from 0.0% to 11.1%. These are descriptive ranges, not evidence of a stable population ranking. Floor/ceiling patterns motivate calibrated difficulty and local metrics before a larger comparison.

5.2. Paired Tools, Representations, and Training

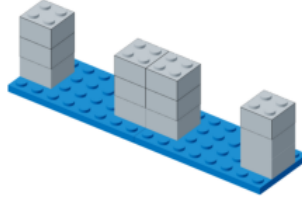
A preceding 24-source, two-model study contains 674 one-shot calls and 56 paired controls. It is not pooled with the three-source matrix. Both models fail exact ordinary reconstruction while succeeding on all recolor/removal cases. Eight-source Gemini controls reuse the same first answer for reflection and target-blind validation (Table 7). Symbolic reference supplies privileged geometry; its gain is not a pure perception intervention. A zero exact-success floor under missing/mismatched images also cannot establish that images are ignored.

Ten SmolVLM-256M [20] base/LoRA [9] pilot jobs share 337 cases and matched supervision, not compute. No positive edit transfer is observed; base outputs all fail formatting. These and earlier Qwen3 [26] pilots are detailed in the supplement.

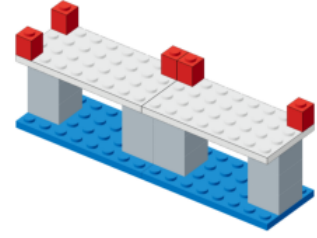
Legal prefix: 7/28 parts



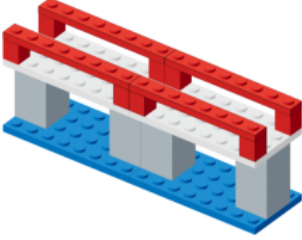
Legal prefix: 14/28 parts



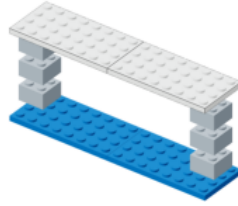
Legal prefix: 21/28 parts



Legal prefix: 28/28 parts



Invalid strategy: decks too early



Blocked inner pier outlined

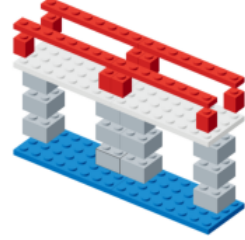


Figure 5. **Ground-truth construction versus a designed dead end.** Four snapshots follow the curated double-span bridge’s verified height-sorted sequence. The last two panels contrast a legal prefix that places both decks before the inner piers with the exploded target; the missing inner pier can no longer descend vertically. The evaluator retains the submitted order and first rejection rather than repairing it.

	Symbolic	RGB	Constraints	RGB	Symbolic	RGB			
Model / algorithm	Relations	Recon.	Gen.	Complete	Edit	Plan	Repair	Format	USD
<i>Vision-language models: same three source objects</i>									
GPT-4.1 mini	3/6	0/3	0/3	0/3	6/9	2/6	4/9	38/39	0.071
GPT-4.1	1/6	0/3	0/3	0/3	6/9	2/6	2/9	35/39	0.341
Gemini 2.5 Flash	2/6	0/3	0/3	0/3	6/9	2/6	3/9	34/39	0.089
Qwen3-VL-32B	0/6	0/3	0/3	0/3	6/9	0/6	1/9	22/39	0.025
Claude Haiku 4.5 ⁺	0/6	0/3	0/3	0/3	6/9	0/6	2/9	22/39	0.317
Gemini 3 Flash (preview) ⁺	3/6	0/3	1/3	0/3	6/9	4/6	1/9	27/39	0.144
Qwen3-VL-8B ⁺	1/6	0/3	0/3	0/3	6/9	0/6	1/9	26/39	0.025
Llama 4 Maverick ⁺	0/6	0/3	0/3	0/3	5/9	1/6	1/9	39/39	0.037
<i>Public-input algorithmic controls: supported tasks only</i>									
Copy current	–	–	–	0/3	0/9	–	3/9	–	–
BOM / extent retrieval	–	0/3	3/3	0/3	–	–	0/9	–	–
Symbolic rules	6/6	–	–	–	9/9	–	–	–	–
Greedy legal action	–	–	–	–	–	5/6	–	–	–
Height-sorted plan	–	–	–	–	–	6/6	–	–	–

Table 5. **Main descriptive matrix: eight models, identical three-source tasks.** Entries are exact successes / cases; variants share sources, so denominators are not independent trials. ⁺ denotes the four newly run models. Algorithm rows use exactly the same cases on supported tasks; – means unsupported or no API charge field, not zero success. No pooled rank or significance claim is supported by three objects. Recognition is separate in the 24-source study because this shared screen has no isolated-part cases.

5.3. Candidate Consistency and a Visual Control

Coordinate-normalized two-piece probes compare full reconstruction, local pose, candidate selection, and privileged symbolic answers. A 12-source, four-model follow-up separates reordering from repetition. For selected poses

a_g, a_g^r, a_g^π , the paired endpoint is

$$D_g = \mathbf{1}[a_g^\pi \neq a_g] - \mathbf{1}[a_g^r \neq a_g]. \quad (5)$$

Mean excess mismatch is 0.50–0.75 across models, with all 192 requests retained. Source-bootstrap intervals [3] are in the supplement. Mismatch is inconsistency, not incorrect-

Model	Reconstruction		Completion		Planning	Repair	
	Part F1	Surface F1	Added F1	Preserve	Legal prefix	Fault F1	Truncated
GPT-4.1 mini	4.3	18.6	0.0	98.5	49.3	33.3	0/39
GPT-4.1	2.4	18.2	0.0	100.0	51.8	16.7	0/39
Gemini 2.5 Flash	5.8	29.0	11.1	66.7	44.3	27.8	1/39
Qwen3-VL-32B	0.0	0.0	0.0	33.3	1.4	0.0	3/39
Claude Haiku 4.5	1.1	16.1	0.0	100.0	19.7	0.0	2/39
Gemini 3 Flash (preview)	24.7	35.4	11.1	66.7	78.2	16.7	0/39
Qwen3-VL-8B	0.0	0.0	0.0	33.3	0.0	0.0	3/39
Llama 4 Maverick	2.4	15.9	0.0	100.0	39.6	16.7	0/39

Table 6. Task-specific diagnostics on the same three sources, percentages except truncation counts. All returned responses remain in each applicable mean, including format failures. Fault F1 includes only the six faulty repair cases, excluding the three normal controls.

Paired condition	Original	Intervention
no-image	0/8	0/8
mismatched-images	0/8	0/8
symbolic-reference	0/8	8/8
reflection	0/8	0/8
validation	0/8	0/8
repeat	0/8	0/8
relative	6/8	0/8

Table 7. Measured eight-source Gemini controls. Relative denotes assembly representation; other rows are reconstruction. These are a different cohort from Table 5.

ness.

A known-renderer mask-matching control uses public BOM and candidates, not target geometry. It yields 44/44 image-conditioned successes and 16/16 no-image abstentions. This is a signal control, not a general reconstructor.

6. Experimental Coverage and Remaining Work

Diagnostic-2’s public symbolic control passes 36 maintenance cases; a known-renderer color control passes their 36 layer-image counterparts. It reads public pixels/current geometry, not target JSON, and is not a general pose estimator. Constructibility-1 public solvers pass all 78 conditions. Always-valid, no-operation, unavailable-part reuse and always-h0 controls pass 6/18, 0/18, 0/6 and 5/36 respectively; the first six are the valid sequence controls. These checks establish executable, nonconstant task signals, not model discrimination. Independent objects, human review and matched-budget model experiments remain necessary.

7. Limitations and Conclusion

BrickAtlas is currently a controlled grid benchmark, not a comprehensive semantic or physically validated assembly benchmark. V2 retains its common base, 19 assembled

types, suffix-only completion, topmost repairs, and height-solvable planning. Frontier raises size to 408 parts, but retains the 25-type rectangular vocabulary and has no learned-model measurements. Constructibility-1 adds per-prefix support and a one-stud side-clearance proxy, not force balance, grasping, or continuous robot motion. Its local roll-back and stockout tiling remain exactly solvable; size alone does not prove difficult reasoning or empirical discrimination. The ambiguity extension is exhaustive only within its stated grammar; nine parameter families are not independent external sources. Pixel equality is verified on one renderer, not every camera or human observer. Current human-review packets contain no real submitted judgments. Small adaptive model studies and same-machine replay do not replace independent confirmation and second-environment reproduction.

Within this scope, the benchmark makes useful distinctions executable: changed-region success is not global similarity [6]; an admissible structure does not determine every hidden relation; and final validity does not certify prefix support, tool clearance, recovery, or calibrated commitment. Stronger claims require independent licensed sources, matched visual process inputs, learned-model and human evidence, and physical validation.

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